

BLACK BOX GENERAL MANUAL – EXTENDED 2 CAN



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1. Black Box overview – Extended

1.1 Technical specifications:

Parameter	Details
Power Supply	6 to 50VDC
Power consumpt.	Down to 100µA
Antennas	Externally
Connectivity	LTE-M, LoRa, BLE
LoRa Technology	868MHz / 915 MHz, up to 1km
Sensors	3D accelerometer (16g), baro-/altimeter (±10 cm)

Component	Specification
Modem	Quectel BG95-M3 LTE Cat M1 / NB-2, GSM Module, all global certifications and R&TTE directives
Frequency Bands	GSM/GPRS: 850/900/1800/1900 MHz LTE: B1-5, 8, 12, 13, 14 (Cat M1) 18, 19, 20, 25, 26, 27 (Cat M1), 28
LoRa	Semtech SX1272 transceiver, Frequency: 868/915 MHz, Protocol: LoRaWAN 1.0.2 and custom protocol, Transmitting power: up to +20 dBm, Sensitivity: -137 dBm
RF Communication	Nordic nRF51822, Frequency: 2.45 GHz, Protocol: BLE 4.0 and custom 2.4 GHz protocol, Transmitting power: up to +20 dBm (with on-board amplifier), Sensitivity: -93 dBm (BLE)
GPS Receiver	Glonaass/GPS/Galileo/Beidou/QZSS - LC76G module
GPS Frequency	GPS L1 1575.42 MHz. C/A Code, 48 search channels Glonaass L1 1598.0625 ~ 1605.375 C/A Code
GPS Sensitivity	Acquisition: -148 dBm Reacquisition: -160 dBm Tracking: -165 dBm
GPS Accuracy	Horizontal Position Accuracy: <2.5 m CEP
Operating Temperature	-25°C to +85°C (without LiPo battery), -20°C to +60°C (with rechargeable LiPo battery)
Power Supply	Maximum range +6 to +31VDC
Charging Current	Max 450mA, observing 0...+45°C safety range for LiPolymer
Power Consumption	20 mA, GPS full power tracking, open GPRS session 100mA BLE transmissions 100 uA, GPS/GPRS/sensors power down, 4 inputs and 1 timer active

1.2 Connections layout:

On the front side of the box the Antennas connections are located. For our installations we will be **using GPS and GSM antennas**. Other connections might be used for future developments.

Both, cables and connections are marked with GPS & GSM respectively. Be sure to do not invert the connections during installation.



LoRa	External LoRa antenna
RF	External 2.45GHz. antenna
GPS	External GPS antenna
GSM	External GSM/GPRS antenna

There are two antenna types which the BU's will be able to request from HQ.

- Internal antenna. This will be installed inside of a “dry” compartment. Non IP-67 antenna. Antenna should be installed on a window or below glass/plastic surfaces. Ex. Operator cabin – Carrier installations or inside of a crane operator cabinet.
- External Puk antenna. This will be installed outside of an electrical cabinet. Weather proof antenna. Ex- BB installed inside of a metal electrical cabinet. The installer must drill a hole of xx mm diameter where the antenna will be fitted. Refer to installation chapter.

On the back side of the black box the I/O's, supplies and can connections are located. Please see below a description of all the pins. For our application only **the bolded pins will be used**. Information about the connections can be found on the connection sheet (single or double engine) or – if available - on the specific installation guide from the crane type.



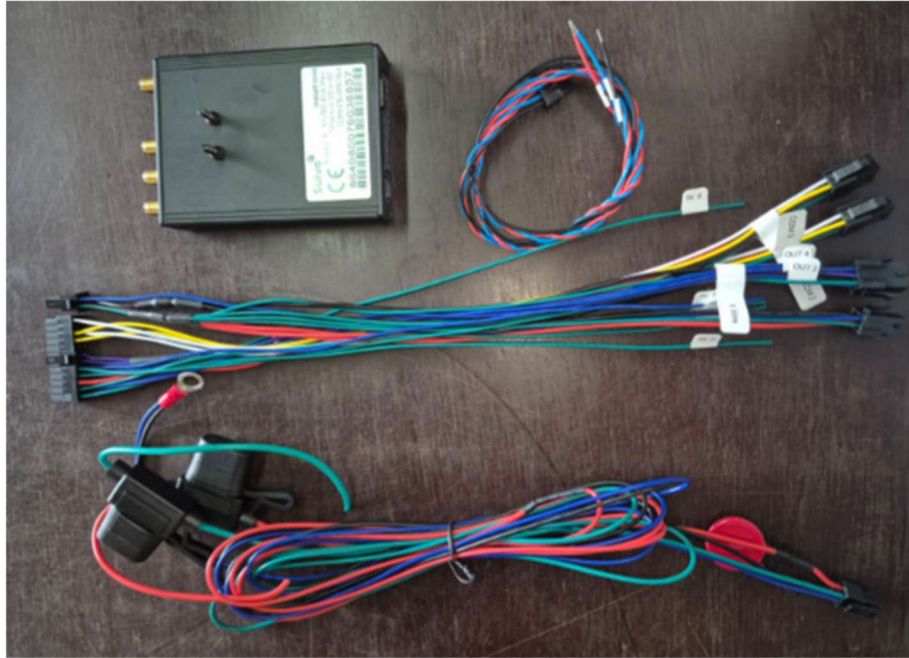
Pin	Signal	Description
1	GND for VCC	Ground for VCC – Chassis ground
2	VCC	+6...+31VDC or VCC Charge input – Battery supply

3	GND for I/O	Ground for I/O
4	In5	Digital/Analog Input 5 (0..31V) – Ignition signal
5	TXD1_3V	3 Volt serial transmit port 1
6	RXD1_3V	3 Volt serial receive port 1, hardware pulse counter
7	TXD2_RS232	RS232 serial transmit port 2
8	TXD2_3V	3 Volt serial transmit port 2
9	GND for I/O	Ground for I/O
10	1Wire	1Wire for RFID
11	1Wire	1Wire for RFID
12	Digital_Out1	Open Collector max. 31V /160 mA, protected via Polyswitch fuse
13	Digital_Out2	Open Collector max. 31V /160 mA, protected via Polyswitch fuse
14	GND for I/O	Ground for I/O
15	Digital_Out3	Open Collector max. 31V /160 mA, protected via Polyswitch fuse
16	Digital_Out4	Open Collector max. 31V /160 mA, protected via Polyswitch fuse
17	TXD3_RS232	RS232 serial transmit port 3
18	TXD3_3V	3 Volt serial transmit output 3
19	RXD3_RS232	RS232 receive input 3
20	RXD2_RS232	RS232 receive input 2
21	VCC_3V3	External Supply 3.3V switchable by module
22	N/C	x
23	N/C	x
24	Digital/Analog_In1	Digital/Analog Input 1 (0..31V)
25	Digital/Analog_In2	Digital/Analog Input 2 (0..31V)
26	N/C	x
27	Digital/Analog_In3	Digital/Analog Input 3 (0..31V)
28	Digital/Analog_In4	Digital/Analog Input 4 (0..31V)
E	Can 1 - CAN_H	CAN H
F	Can 1 - CAN_L	CAN L
G	Can 1 - GND for CAN	Ground for CAN – Same as Chassis ground
H	Can 2 - CAN_H	CAN H
I	Can 2 - CAN_L	CAN L
J	Can 2 - GND for CAN	Ground for CAN – Same as Chassis ground

2. Package content – Extended

Following materials will be delivered from HQ:

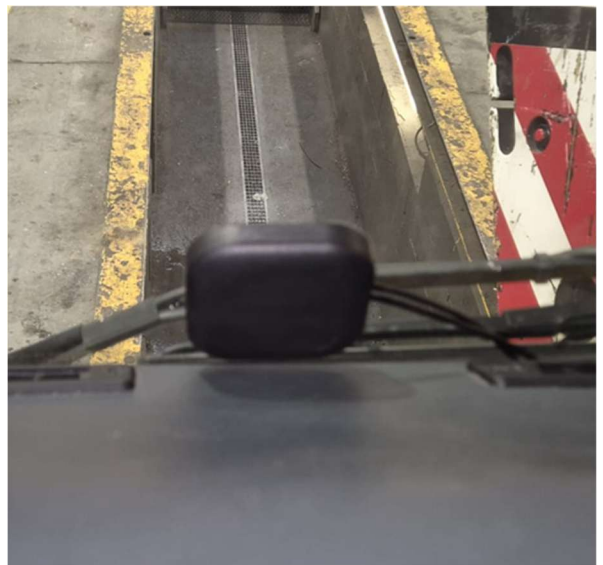
- Extended Black box – sap 10094697.
- Set of wiring – included on sap 10094697 extended BB:
 - o 2x CANBUS harness, I/O harness and supply harness.



- Antenna – BU to decide which antenna to use depending on the installation of the BB – see installation chapter:
 - o Internal – sap 10094701



- External Puck – sap 10094699



3. Tools and requirements

Standard technician – electrician tools are needed to perform the installation, the manual wont specified which set of standard tools the installer should use, but give an overview of the “non common” tools which are required.

- Set of electrical pin extractor tools. Although in most cases this will be a standard kit for any BU, it might defer from a standard tool. For hooking/joining the signal cables from the black boxes to the crane wiring.
- P-can viewer usb dongle – sap 10067404. Dongle used for the installation software tool – See installation check chapter.
- Cable set for P-can viewer – Sap . Set of wires for the p-can viewer tool – See installation check chapter.



4. Pre-installation check.

Similar to our previous Black boxes suppliers, the installation will require some kind of electrical schemas understanding knowledge to identify where the signal of the black box should be hooked.

Before starting the installation, the technician/installer should make sure that:

- all the require tools are available (both hardware and software).
- he should identify in which type of crane is going to install the BB (year of production, manufacture and if an installation example is available on sharepoint and/or the installation tool)

- How many engines the crane has installed and where the BB require signals can be fetched from (Engine running, Upper crane active, SLI override key, crawler).
- Electrical schemas are available.
- Connection points are clear.

4.1 In general and without taking in account the crane manufacture, we can divided the type of crane on:

- **Single engine cranes - All terrain / RT:**
 - o Prior 2012 production. Signals for BB engine running carrier and crane/upper ON most probably coming from normal I/O – 24v signals.
 - o Above 2012 production. Signals for BB engine running carrier and crane/upper ON most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 connected to CanBus crane control.
 - o Signals for SLI override key will come from CanBus only on newer type of cranes – Grove using SLI CanBus on older types.
- **Double engine cranes – All terrain:**
 - o One BB per engine.
 - o Carrier:
 - Prior 2012 production. Signals for BB engine running ON most probably coming from normal I/O – 24v signals.
 - Above 2012 production. Signals for BB engine running most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 Not connected)
 - o Crane/upper:
 - Prior 2012 production. Signals for BB engine running ON most probably coming from normal I/O – 24v signals.
 - Prior 2012 production. Signal for SLI override key coming from normal I/O – 24v signal.
 - Above 2012 production. Signals for BB engine running most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 connected to Canbus crane control)
- **Single engine – Lattice boom cranes (LBC):**
 - o Prior 2012 production. Signals for BB engine running ON most probably coming from normal I/O – 24v signals.
 - o Above 2012 production. Signals for BB engine running most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 Not connected)
 - o Signal for SLI override key coming from normal I/O – 24v signal.
 - o Signal for Crawler coming from normal I/O – 24v signal.
- **Double engine – Lattice boom cranes (LBC):**
 - o One BB per engine.
 - o Engine 1_BB nr 1:
 - Prior 2012 production. Signals for BB engine running ON most probably coming from normal I/O – 24v signals.

- Above 2012 production. Signals for BB engine running most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 Not connected)
- Engine 2_BB nr 2:
 - Prior 2012 production. Signals for BB engine running ON most probably coming from normal I/O – 24v signals.
 - Above 2012 production. Signals for BB engine running most probably coming from CanBus (BB-Can 1 connected to powertrain canbus // BB-Can 2 Not connected)
 - Signal for SLI override key coming from normal I/O – 24v signal.
 - Signal for Crawler coming from normal I/O – 24v signal.
- *** Liebherr crane as from 2018 ***. As from 2018 production Liebherr cranes are equipped with a FMS connector (Fleet management system). This simplifies the connection, allowing all signals needed for our Black box being fetch from the same plug.
 - Single BB
 - BB-Can1 connected to the FMS connector.

4.2 See the below the description of the BB wiring ports-signals and – with a general view – where the wiring should be connected – Referring only to the signals we need for our installation:

- **Pin 1 – Pin 3 – Pin G – Pin I → Ground.**
 - Those are chassis grounds, can be joined together and bolted to a chassis ground. Make sure that a good ground (low resistance to battery ground) has been chosen.
- **Pin 2 → Battery supply.**
 - If the crane is equipped with a battery disconnecter – master switch, the battery supply for the BB it should be connected before.
- **Pin 4 → I/O type. Ignition Signal.**
 - Signal needed to wake up the BB.
 - Single engine crane – Ignition signal from Carrier side.
- **Pin 24 → I/O type Din_1.**
 - * Optional If canbus not available → Engine running signal D+.
- **Pin 25 → I/O type Din_2. SLI – Override key signal.**
 - **Optional*. Only connected if NO Crane CanBus available! See point 4.1.
- **Pin 27 → I/O type Din_3. Crawler movement.**
 - Only on LBC.
- **Pin 28 → I/O type Din_4.**
 - Single engine crane: Ignition signal from Upper cabin_2.

- **CanBus plug 1:**
 - o Pin E → Can H.
 - o Pin F → Can L.
- **CanBus plug 2:**
 - o Pin E → Can H.
 - o Pin F → Can L.

5. SarTel_installation tool.

The installation tool has been developed with the intention of providing a tool to the technicians/installers to check:

- CanBus connection to the BB done correctly.
- Checking baudrate.
- Check the Signals for Engine running, PTO engage (crane operation ON-OFF) and SLI key override.
- Consult BB installation manuals and installation examples.

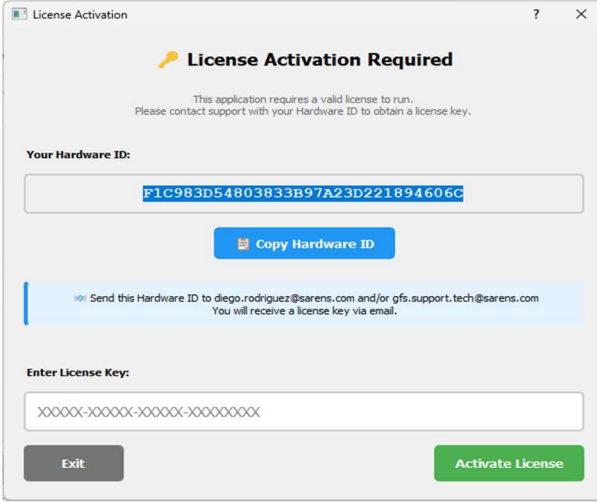
5.1 Hardware requirement:

- Computer/ Windows base tablet. Any pc can be used. The computer should have the software tool and the drivers installed (P-can drivers – Available in Sharepoint > Blackbox > installation tool.
- Pcan USB to Can dongle. Available via HQ or online. Sap 10067404 – See chapter 3 Tools and requirements.
- Set of wires. Available via HQ or self-made (Cross wires with one end a female DB9 connector and other end loose pins.

5.2 Software tutorial:

The first time the tool is used, it will ask for a activation key. Press the button “Copy hardware ID” and contact xxxx@sarens.com to get the activation key. See illustration below.

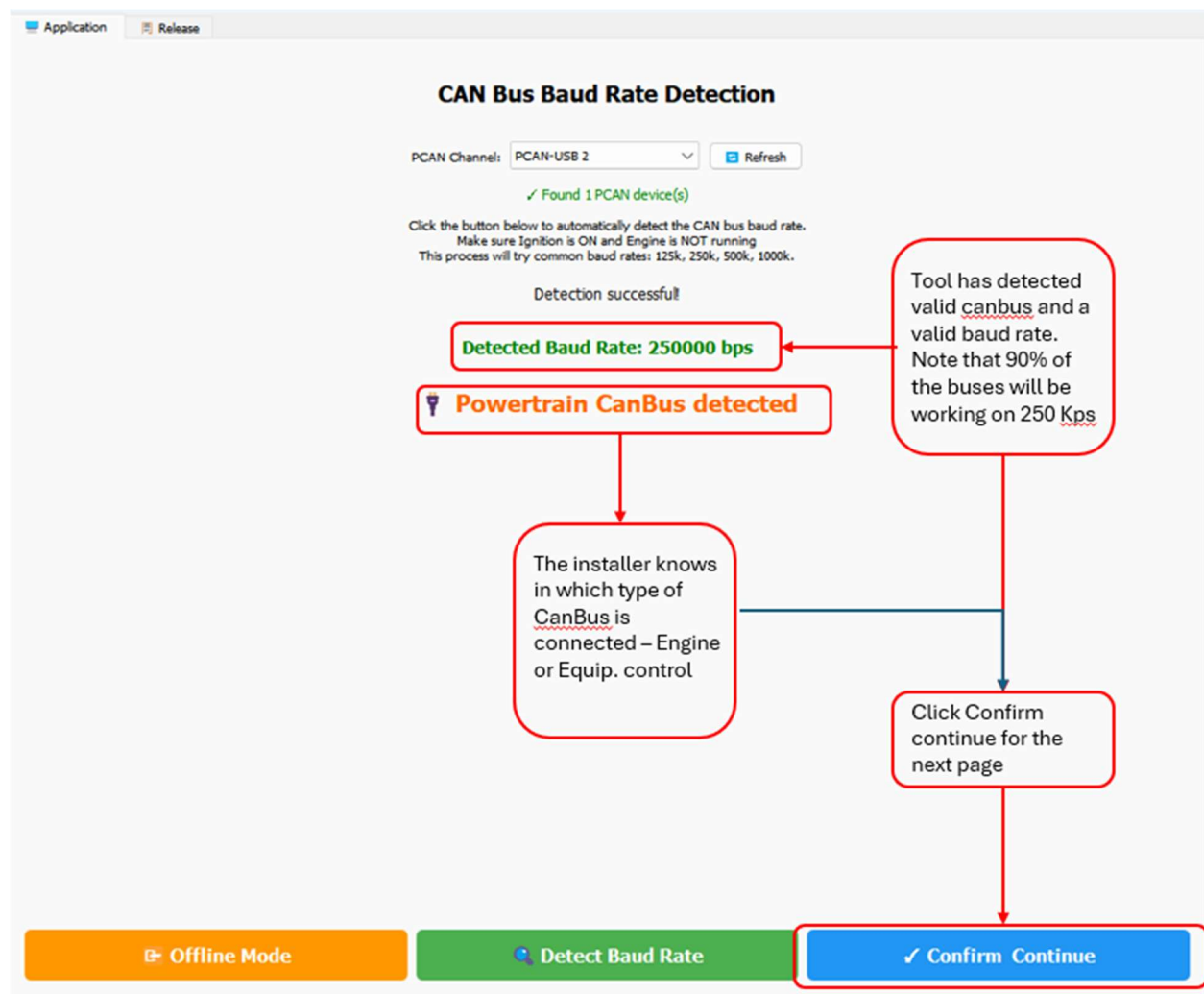
Once you get the license key, copy it below License Key and press “Activate license”.



The screenshot shows a window titled "License Activation" with a close button (X) in the top right corner. The main heading is "License Activation Required" with a key icon. Below this, a message states: "This application requires a valid license to run. Please contact support with your Hardware ID to obtain a license key." The "Your Hardware ID:" is displayed as "F1C983D54803833B97A23D221894606C" in a text box. Below the text box is a blue button labeled "Copy Hardware ID". A light blue instruction box says: "Send this Hardware ID to diego.rodriguez@sarens.com and/or gfs.support.tech@sarens.com. You will receive a license key via email." Below this, the "Enter License Key:" section has a text box with a placeholder "XXXXX-XXXXX-XXXXX-XXXXXXX". At the bottom, there are two buttons: "Exit" (grey) and "Activate License" (green).

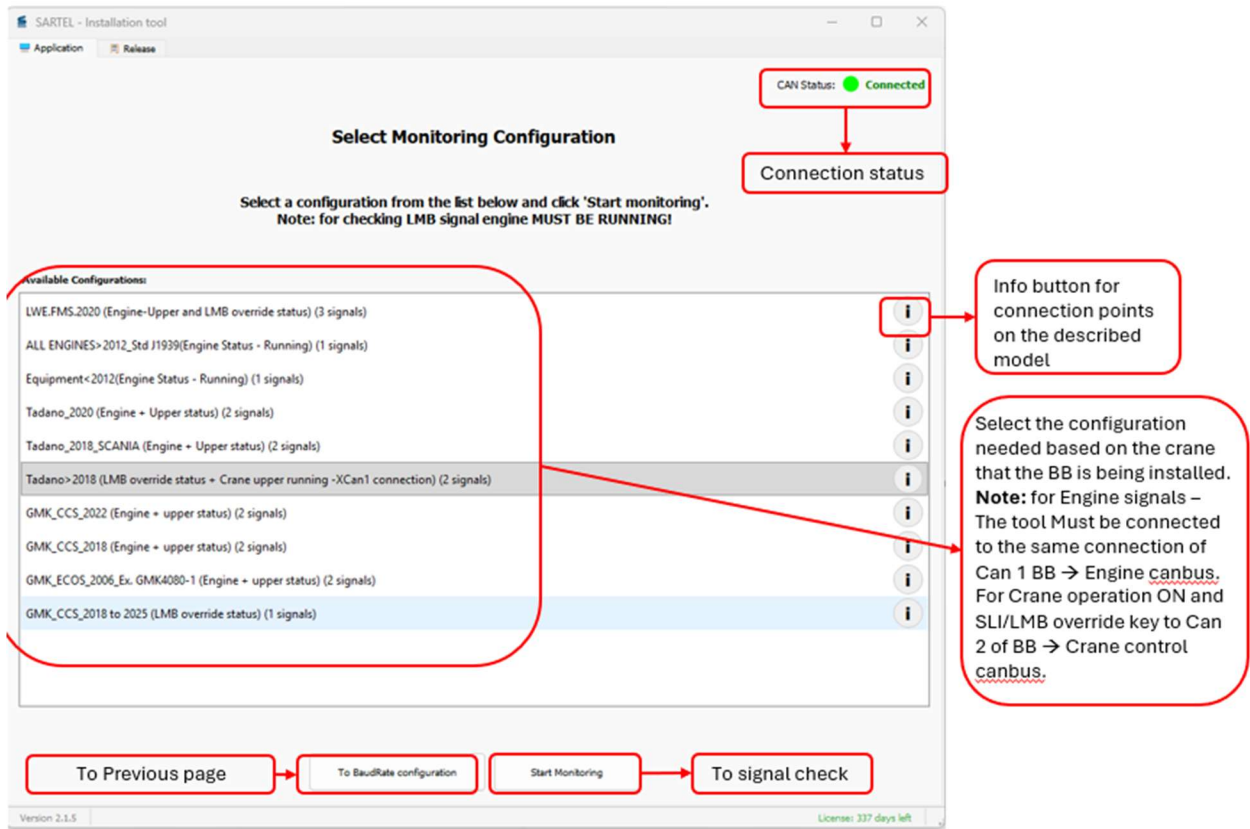
The first page of the tool the user will be able to:

- Select the USB dongle. By default the tool will recognize the P-can dongle, if more than one is connected to the pc, the user can select which one is going to be used.
- Offline mode. The tool can also be used to read documentation and schemas from the installation of different models without being connected to neither the crane or the P-can dongle.
- Detect Baud Rate. Once the P-can dongle is connected to the PC, and the wires from the dongle are connected to the crane, the user can press this button and the tool will scan the canbus (ignition must be switched ON!). If the tool detects canbus, a message with the baud rate will appear on the tool.
- Confirm Continue. After the tool has detected the baud rate and valid canbus, the user can press Confirm to continue to the following page.
- CanBus type identification. The tool will give an indication if the installer is connected to an Engine bus or an equipment control bus

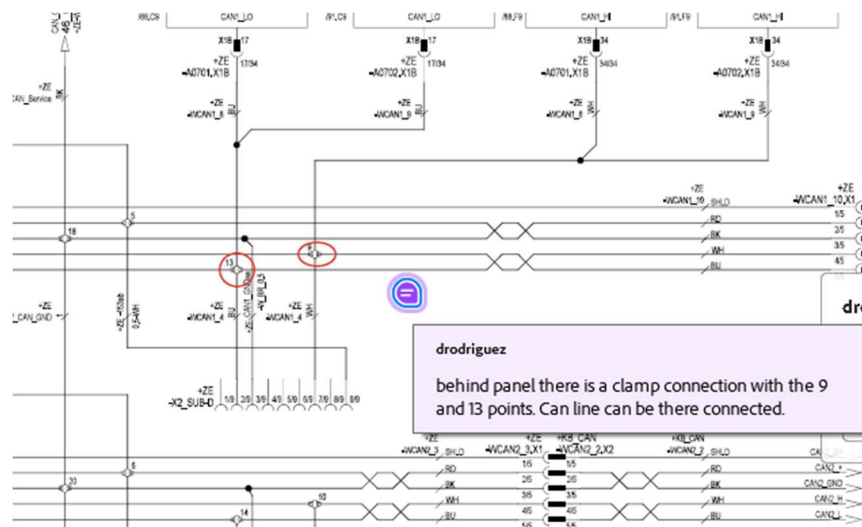


In the second page the installer will be able to:

- Access the connections points based on crane model by pressing the “i” button.
- Select “Start Monitoring” to check if the signals of: Engine running – PTO/ crane operation – SLI/LMB override key activation.

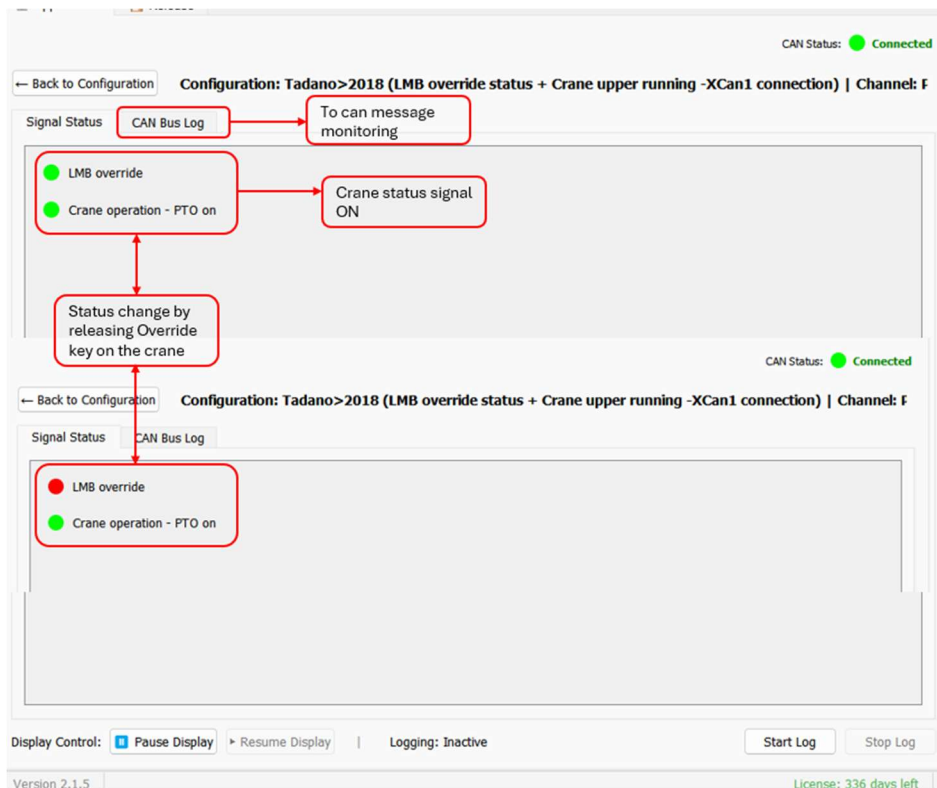
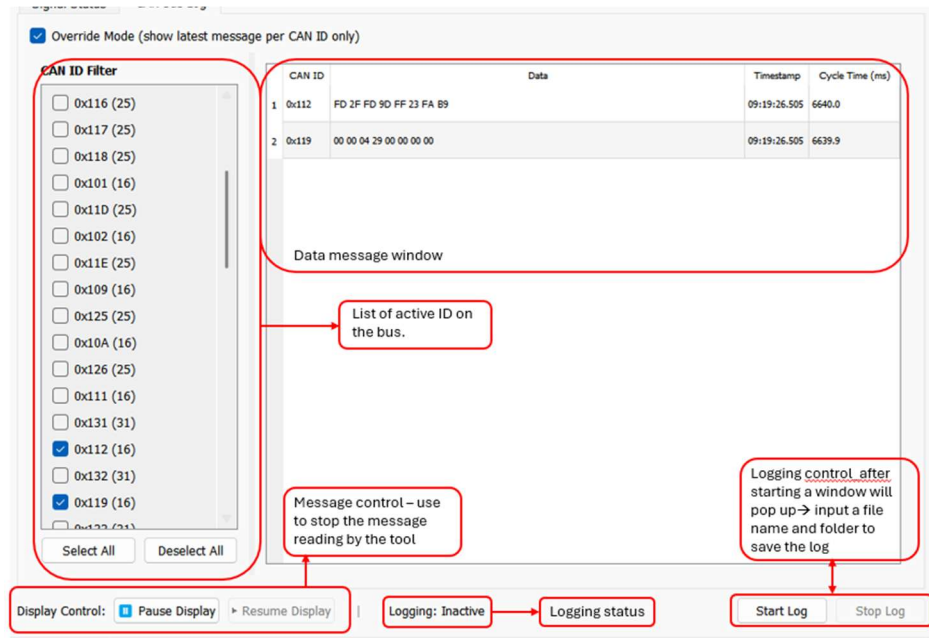


- Example of “i” page.



In the third installer tool page, installer will be able to:

- Check if the signals are becoming active by starting the engine (if on an engine configuration) or checking if crane operation active // SLI/LMB Override key ON (if on a crane control configuration).
- Monitor the messages which are coming in the bus and create a log file which can be sent to Sarens HQ in case of installation issues.



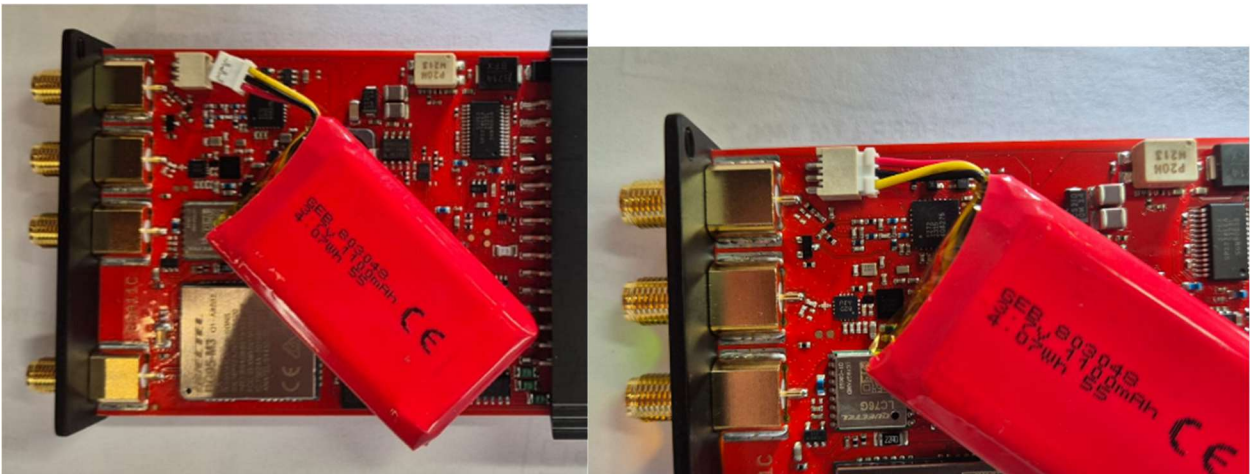
6. Hardware Installation.

As the best way to explain an installation is to use an example, the below instructions refer to 5.1 Tadano crane single engine, 5.2 Demag crane prior 2012 double engine and 5.3 Liebherr crane as from 2018. Further installation example will be included on Sharepoint > installation examples.

Note:

The Black boxes are delivered with the internal battery disconnected – That way the BB can be stored for longer on a warehouse without getting the battery empty.

Prior starting the installation, the installer must connect it:

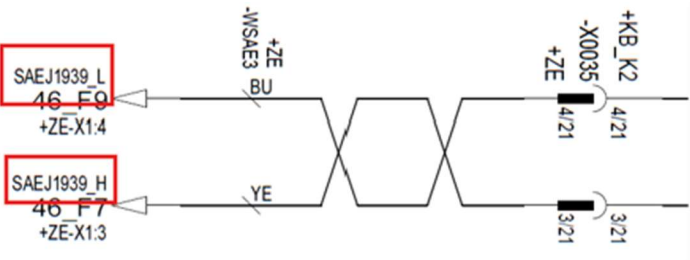


Fit the screws after plugging the battery

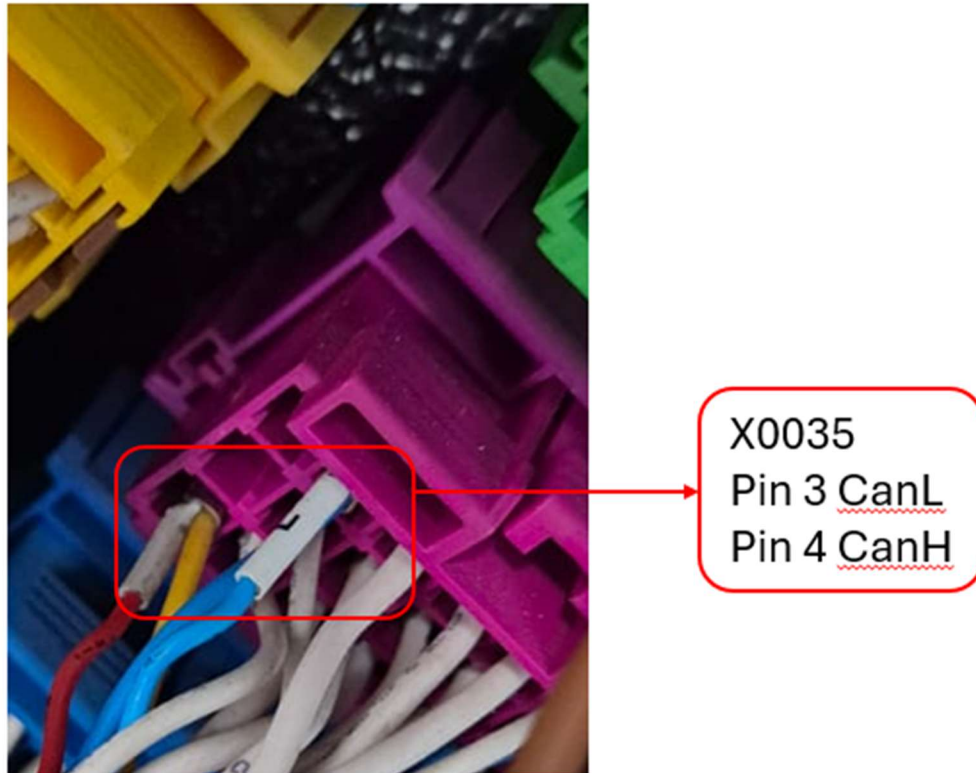
6.1 Installation on Single engine crane from 2022. This crane uses a engine to drive the carrier and also to drive the hydraulic pumps for Crane operations.

For engine data (engine on – Engine hours – engine errors): The installer should take the electrical schemes and identify the Powertrain canbus. In general, canbus wiring can be identified on the schemas by crosswire lines, and

15	IGNITION CAB SUPERSTRUCTURE
16	CAN BUS DISPLAY
17	WFAR INDICATOR HYDRAULIC OIL . ACC.



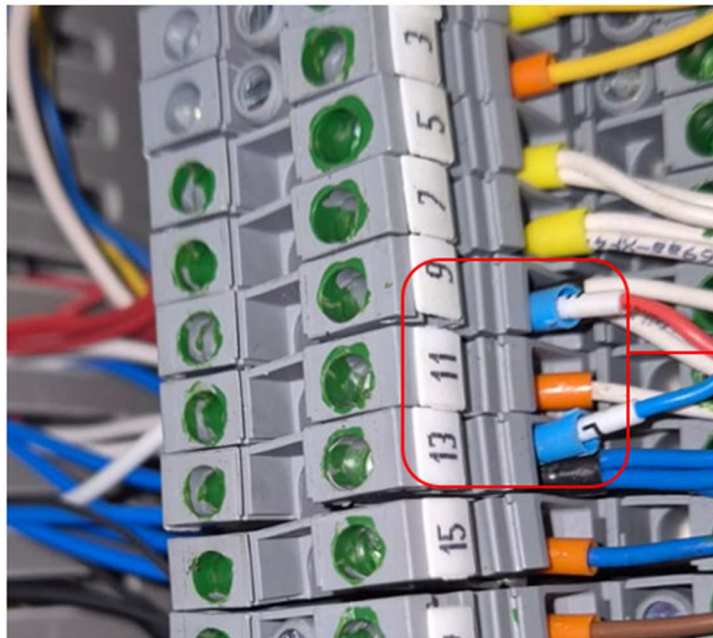
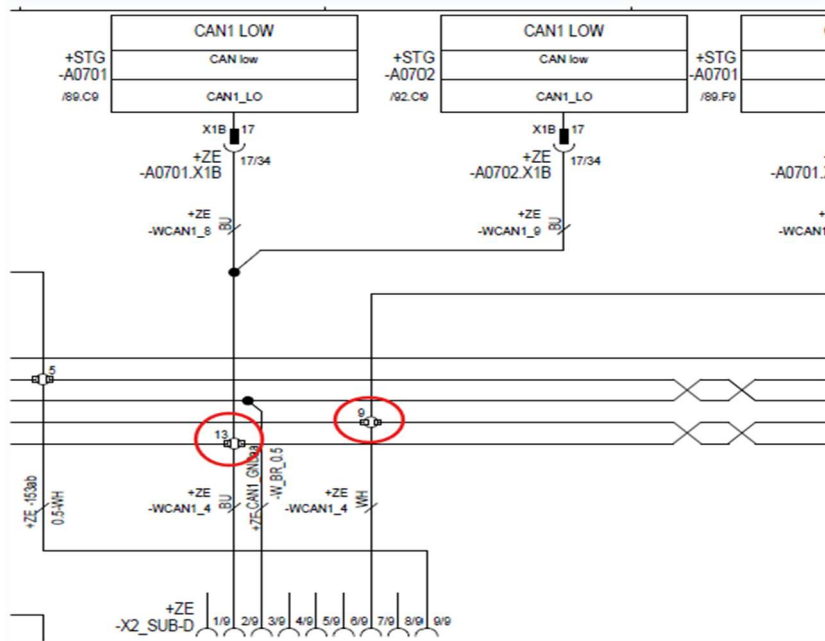
wording as J1939¹. If the crane is using a single engine to drive carrier and hydraulic pump for Upper/crane working, 90% of the cases the engine bus will be available on the upper Structure (US).



In this case we can see that on the US electrical cabinet there is a connector called X0035 where pin 4/21 is Can L (marked as **SAEJ1939_L**) and pin 3/21 is Can H (marked as **SAEJ1939_H**). On this plug the can connections from the BB can be connected (Can H BB to Can H crane // Can L BB to Can L Crane).

¹ J1939 is a CanBus standardized engine communication protocol broadly used for engine manufactures since early 2000's. In most heavy vehicles this protocol offers a standardized language define by SAE international, although developed from early 2000 stages, some manufactures only implemented as from around 2010 and depending on the engine capacity. As a rule of thumb we will consider an crane equipped with an engine using J1939 as from 2011/12. This standardized protocol broadcast information as Engine hours, Engine fuel consumption, Oil pressure, water level, error codes...and many more. Using a BB with those capabilities will increase our future efficiency.

For Crane control canbus the principle is the same, to identify canbus wiring on the schemes the principle remains → CanBus wiring is drawn as cross wired cables. Refer to the Crane specific installation manual to know which canbus wiring should be chosen.



Wcan1_4
Pin 13 CanL
Pin 9 CanH

Note Canbus wiring can be physically identified on the equipment by crosswired cable often with following color combinations: Blue-white // Green-yellow // Orange-green

For Battery supply (Pin 2), ground (Pin 1-3-E-G-I) and ignition signal, the installation is the same as previously done for Multicap-Globetracking BB's.

- Battery supply. Switch off the battery disconnecter (if the crane is equipped with it) and with a multimeter look for +24v.
- Chassis ground. Connect the pins to a chassis ground.
- Ignition signal. Identify a connection point where there is +24v when the ignition key is ON. This signal is important to wake up the box and start the data recording among other things. As in most of the cases the signal won't be the same when carrier ignition is on and crane ignition is on, it should be connected:
 - o Carrier ignition: Din 5 (supply connector pin 4)
 - o Crane ignition: Din 4 (I/O connector pin 24) ***only needed for single engine cranes.**

From both engine CanBus and crane CanBus all signals for engine running, crane operation and SLI override will be fetched.

6.2 Installation on a double-engine crane from 2010. This equipment is using an engine to drive carrier-lower structure hydraulics and another engine to drive crane hydraulics. **Note: There will be always a BB installed per engine.**

In certain cases engine from this age can also be transmitting J1939 canbus data – Check example installation sheets for specific cases. Generally we will only use the CanBus capabilities on equipment produced as from 2011.

Supplies connection remains the same as previous examples or Multicap boxes:

- For Carrier engine:
 - o Battery supply. Switch off the battery disconnecter (if the crane is equipped with it) and with a multimeter look for +24v.
 - o Chassis ground. Connect the pins to a chassis ground.
 - o Ignition signal. Identify a connection point where there is +24v when the ignition key is ON. This signal is important to wake up the box and start the data recording among other things.
- For Crane engine/ SLI override:
 - o Battery supply. Switch off the battery disconnecter (if the crane is equipped with it) and with a multimeter look for +24v.
 - o Chassis ground. Connect the pins to a chassis ground.
 - o Ignition signal. Identify a connection point where there is +24v when the ignition key is ON. This signal is important to wake up the box and start the data recording among other things.
 - o SLI override: Connect to the switch of the EN13000 override key.

6.3 Liebherr crane as from 2018. Since 2018 Liebherr integrates in production a FMS connector (Fleet management system). This connector allows us fetching all carrier, engine and crane operation data.

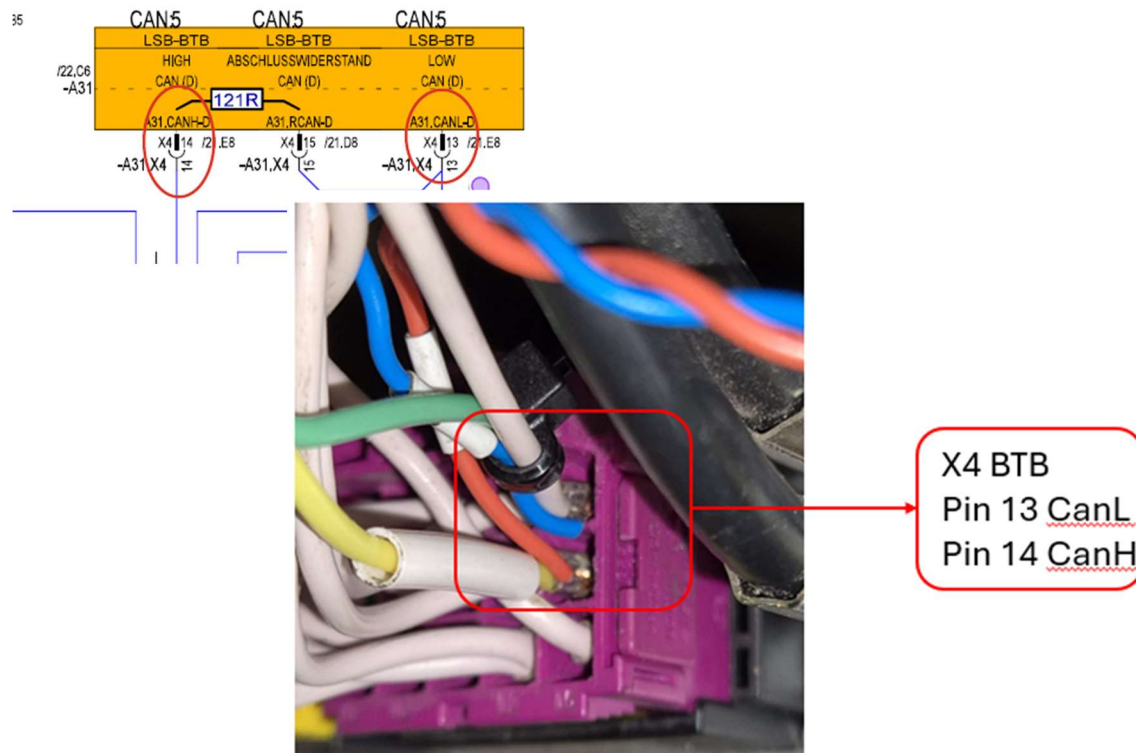
The connection points for this Canbus is located on the BTB's. See BMK information or example installations for more specific information. Generally the BTB is installed on the carrier middle console for LTM cranes, and electrical cabinets for lattice boom cranes (LBC).

Supplies remain the same as previous examples:

- Battery supply. Switch off the battery disconnecter (if the crane is equipped with it) and with a multimeter look for +24v.
- Chassis ground. Connect the pins to a chassis ground.
- Ignition signal. Identify a connection point where there is +24v when the ignition key is ON. This signal is important to wake up the box and start the data recording among other things.

Can Connection – when FMS is available there will be a yellow and green cables on the plug X4 – pin 14 (Can H) and pin 13 (Can L).

Connect Can wires from the BB (Can L to 13 and Can H to 14) as shown on the pictures below.



7. Installation sheet.

At the end of the installation the technician/installer must fill up the installation sheet (templates available in Sharepoint), take the required pictures – as specified in the installation sheet and forward a copy to xxxx@sarens.com.

The installation sheet will be uploaded to SarPlan under the crane resource number.

There are three different installation sheets depending in which equipment the BB is installed:

- Single engine crane – 1x Extended 2 can BB
- Double engine crane – 2x Extended 2 can BB
- MHE/Pool cars/Truck – Pulse BB
- Trailers – Trailer BB

The example below shows the explanation of the most complete BB template sheet – Single engine-Extended 2 Can.

In any case, the responsible fleet manager/fleet team should provide the crane documentation, BB general installation manual, crane specific installation example (if available) and all necessary tools to perform the installation on a professional way.

The installer is responsible for reading and understanding the provided documentation, checking the functionality of the BB with the installation tool, filling up correctly the installation sheet and provide the required pictures.

The first part of the connection sheet is taken over our legacy Multicap system, the installer should fill up the required information of the crane, BB IMEI, MO,

Crane number	Brand /Type	Date	Supplier
		Technician	SUIVO / GFS
Serial number crane	IMEI number Black Box	Maintenance order (SAP)	

The supplies connection points are basically the same for all the BB types (and previous legacy boxes). This part of the connection sheet is self-explanatory.

Connection for supplies:								
	Pin	Naming	Data type	Info page schemas	Wire color	Crane connector name (From schemas)	Signal checks	Signal function on Platform
	2	24V	KL 30 - Battery supply				Supply from battery KL 30+. Before master switch! Write voltage value:	Power supply for the tracker when the machine is not in used
	1	Ground	KL 31 - Chassis ground				Ground / masa connection	Ground
	3	Ground	KL 31 - Chassis ground				Ground / masa connection	Ground
	4	Ignition signal	KL 15 - Ignition signal				Ignition signal KL+15. Note that BB needs to have ignition signal for data collection. Write voltage value:	Wake up signal for CanBus data collection -  The Canbus data recording will start when the signal is detected. Ex. Single engine- The BB is installed on the US, the ignition signal used for BB should be high when the carrier or crane ignition are on (see crane connection examples)

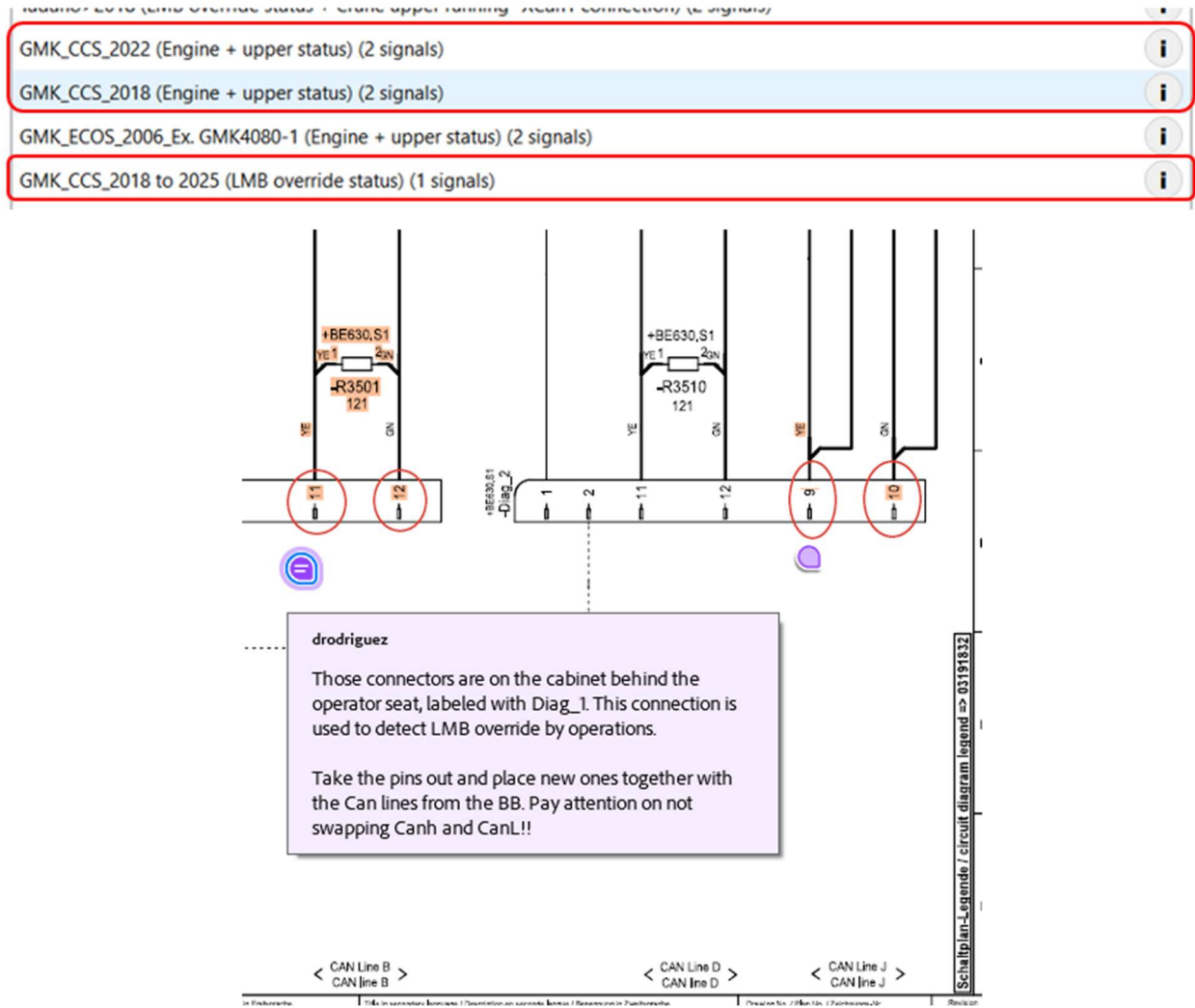
On the connection for data collection the installer should check on which type of crane he is installing the BB.

- Is it a Liebherr as from 2018? Does it has an FMS connection?
- Is it a crawler or All terrain crane?
- Which crane manufacturer and year of production is the crane?

- Is there a installation example available?

Base on that it have to choose the installation method.

Example 1: Grove GMK 5250L from 2017. In this crane (above 2012) there is engine data available and on the installation tool we can see a configuration option.



The technician/installer should double check the example sheet with the crane schemas and proceed with the physical installation and check with the installer tool.

Once done, the installation sheet can be filled. As the connection has been done for engine, LMB/SLI override key and crane/upper operation in canbus, the check boxes should be ticked as shown below:

Other crane brand or Liebherr older than 2018:

CAN 1 or I/O	E	Engine data	CanBus - Connected to the engine can	☒		Can H :	Installation tool - Sartel- check & note Baudrate:
	F	CAN_H				Can L :	Installation tool - Sartel- check & note Configuration:
	24	Digital In 1	I/O standard D+	☐			Installation check with multimeter - Write Voltage value
CAN 2 or I/O	H	SLI - Override key / Crane on	CanBus - Connected to the crane control	☒		Can H :	Installation tool - Sartel- check & note Baudrate:
	I	CAN_H				Can L :	Installation tool - Sartel- check & note Configuration:
	25	Digital In 2	I/O standard from override key	☐			Installation check with multimeter - Write Voltage value

Fill the schema page where the connection is found

Fill the connector number and pin number

Fill the information taken from the installation tool

Information about current working hours and attached the pictures requested.

Information working hours	
Working hours Primary Engine / Lower Engine :	
Working hour Upper:	
Travelling (crawler):	
Specified position of BB in the crane:	
Travelling (crawler):	
Take pictures of the following:	
IMEI sticker:	☒
Location of BB in the crane:	☒
Connection of Can 1:	☒
Connection of Can 2 (if connected):	☐
Engine id plate:	☒

Once completed the installer should forward the pictures and connection sheet. The fleet manager is responsible to send the information to xxx@sarens.com. Once received the Black box team will configure the box over the air, it will then become visible on the platform and data will be uploaded on AX.